Diploma HRI Epidemiology

health records and information technology (Kenya Medical Training College)

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Epidemiology I (30 hours)

Unit 1. Concepts to epidemiology

1.1. Definition of epidemiology

- Branch of medicine which deals with incidence, distribution, and control of diseases and other factors relating to health
- Analysis of the distribution, patterns and determinants of health and disease conditions in defined populations.
- Study of distribution and determinants of health-related states or events, and the application of this study to the control of diseases and other health problems
- Study of how often diseases occur in different groups of people and why

1.2. Origin / history of epidemiology

- Epidemiologic thinking has been traced from Hippocrates
- In 400 B.C, Hippocrates explained that environmental and host factors such as behaviors might influence the development of disease.
- In 1662, John Graunt published analysis of mortality data that quantified patterns of birth, death, and disease occurrence by sex, region and seasonal variations
- In 1800, William Farr collected and analyzed Britain's mortality statistics and developed basic practices that are being used today in vital statistics and disease classification.
- In 1854, John Snow conducted a study when an epidemic of cholera erupted in the Golden Square of London.
- In the 18th and 19th Century, epidemiological methods began to be applied in the investigation of acute infectious diseases.
- In the 1930s and 1940s, epidemiologists extended their methods to noninfectious diseases
- During the 1960s and early 1970s health workers applied epidemiologic methods to eradicate naturally occurring smallpox worldwide
- In the 1980s, epidemiology was extended to the studies of injuries and violence.
- In the 1990s, the related fields of molecular and genetic epidemiology took root.

1.3. Scope of epidemiology

- **Causation** – Describes how diseases are caused by interaction between genetic and environmental factors
- **Natural history of disease** – Explains the course and outcome of a disease in individual or group
- **Health status of the population** – Describes health situation of the population
- **Evaluation of interventions** – Evaluates effectiveness and efficiency of intervention

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1.4. Purposes of epidemiology

- The following are purposes /uses of epidemiology:
    - Count health-related problems
    - Clarification of natural history of disease
    - Describe the distribution of health-related events in the population
    - Describe disease patterns
    - Describes health status of population
    - To determine the progression and extent of disease
    - Identify risk factors for developing diseases
    - Identify causes (etiology) or determinants of disease
    - Identify control and/or preventive measures
    - Establish priorities for allocating resources
    - Select interventions for prevention and control
    - Evaluate effectiveness of programs and interventions
    - Conduct research (descriptive or experimental i.e. drug trials / vaccine trials)

1.5. Principles of epidemiology

- The following are principles of epidemiology:
    - Show distribution with the frequency and pattern of health events in a population.
    - Search for causes and determinants of disease occurrence
    - Concerned with health of people in specified populations
    - Provides data that can be applied for public health action
    - Use of scientific methods to describe and analyze situations
    - Define and count cases to determine their distribution, calculate rates and compare these rates

1.6. Terms used in epidemiology

- **Agents** - are biological, physical, or chemical factors that contribute to the occurrence of a disease e.g. viruses, bacteria, fungi, poisons or allergens, radiation, noise, or heat

- **Host** - A person or living organism that can be infected by an infectious agent under natural conditions

- **Host factor** - An intrinsic factor (age, race, sex, behaviors, etc.) which influences an individual's exposure, susceptibility, or response to a causative agent

- **Environment** – the habitat in which the biological agent and the host exist, survive or originate

- **A vector** is an organism that helps transmit infection from one host to another.

- **Vehicle** - An intermediary in the indirect transmission of an agent that carries the agent from a reservoir to a susceptible host

- **Case** - an individual with the particular disease

- **Incidence** - number of new cases in a population at risk of developing the disease over a given period of time.

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- **Incidence rate** - is the number of new cases divided by the total disease free person-time of observation seen in the population at risk

- **Prevalence** - quantifies the number of existing cases in a population at a point in time or during a period of time

- **Observations** – is one form of data collection through direct seeing or hearing how people act.

- **Endemic** – means the constant presence/occurrence of a disease in a population and the disease is able to maintain itself in a population or region without cases entering the population or region from outside

- **Epidemic** refers to outbreak (sudden increase in the number) of cases of a disease above what is normally expected in that population in that area.

- **Pandemic** - An epidemic occurring over a very wide area (several countries or continents) and usually affecting a large population i.e. Covid-19.

- **Hyperendemic** refers to persistent, high levels of disease occurrence.

- **Holoendemic** - A disease is holoendemic when essentially every individual in a population is infected

- **Carrier** - A person or animal without apparent disease who harbors a specific infectious agent and is capable of transmitting the agent to others.

- **Case control study** - A type of observational analytic study where enrollment into the study is based on presence ("case") or absence ("control") of disease.

- **Case definition** - A set of standard criteria for deciding whether a person has a particular disease or health-related condition.

- **Case fatality rate** - The proportion of persons with a particular condition (cases) who die from that condition

- **Cohort** - A well-defined group of people who have had a common exposure, and are followed up for the incidence of new diseases or events.

- **Cohort study** –starts by defining groups by the study factor (e.g. exposure present or absent) and then follows-up these exposure groups to detect the outcome (e.g. disease present or absent).

- **Determinant** - Any factor that brings change in a health condition

- **Pathogenicity** - The proportion of persons infected, after exposure to a causative agent, who then develop clinical disease

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- **Reservoir** - The habitat in which an infectious agent normally lives, grows and multiplies. I.e. human reservoirs, animals' reservoirs, and environmental reservoirs.

- **Risk** - The probability that an event will occur, e.g. that an individual will become ill or die within a stated period of time or age.

- **Risk factor** - Any personal behavior or lifestyle, environmental exposure, inborn or inherited characteristic that is associated with an increased occurrence of disease or other health-related event or condition.

- **Trend** - A long-term movement or change in frequency, usually upwards or downwards.

- **Virulence** - The proportion of persons with clinical disease, who after becoming infected, become severely ill or die.

- **Zoonosis** - An infectious disease that is transmissible under normal conditions from animals to humans.

- **Odds ratio** - A measure of association which quantifies the relationship between an exposure and health outcome from a comparative study

- **Attributable risk** – difference between the incidence of the outcome in the exposed group (or intervention) and the incidence of the outcome in the unexposed group (or control).

- **The relative risk (RR)** - is the ratio of the probability of an outcome in an exposed group to the probability of an outcome in an unexposed group

- **Absolute risks** - It is the ratio of people who have a medical event compared to all of the people who could have an event.

- **Causality** - identifying the causes of disease

- **Chronic disease** - a long-lasting, persistent or recurrent disease

- **Confounding bias** – occur if the effect of the study factor on the outcome is mixed in the data with the effect of another variable

- **Placebo** – is a substance or procedure with no known effect used as mock treatments for the control group in experimental studies.

- **Positive predictive value** - probability that a person with a positive test result will actually have the disease.

- **Surveillance** - is the ongoing systematic collection, analysis and interpretation of data for planning, implementation, and evaluation.

- **Target population** - is the population about which one wants to draw conclusions.

- **Analytical epidemiology** - concerned with the search for health-related causes and effects by quantifying the association between exposures and outcomes

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- **Applied epidemiology** - The application or practice of epidemiology to address public health issues.
- **Association** - Statistical relationship between two or more events, characteristics, or variables.
- **Active immunity** - Resistance developed in response to stimulus by an antigen
- **Sensitivity** - The ability of a system to truly detect presence of epidemics
- **Specificity** – Ability of a system to truly detect absence of a disease
- **Sporadic** - A disease that occurs infrequently and irregularly
- **Validity** - The degree to which a measurement actually measures or detects what it is supposed to measure.

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Unit 2. Epidemiology and disease transmission

2.1. Causation of disease

- A number of models of disease causation have been proposed. Among the simplest of these is the epidemiologic triad or triangle
- The triad consists of an external agent, a susceptible host, and an environment that brings the host and agent together.
- In this model, disease results from the interaction between the agent and the susceptible host in an environment that supports transmission of the agent from a source to that host

Agent

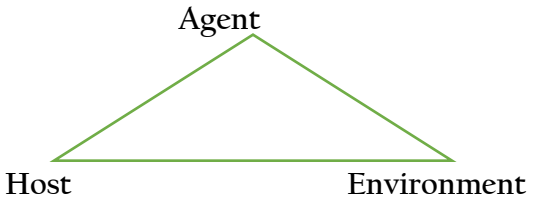
- Is referred to an infectious microorganism or pathogen: a virus, bacterium, parasite, or other microbe.
- Agent factors - These factors must be present for a disease to occur
- The agent must be present for disease to occur; however, presence of that agent alone is not always sufficient to cause disease.
- A variety of factors influence whether exposure to an organism will result in disease, including the organism's pathogenicity (ability to cause disease) and dose.
- Over time, the concept of agent has been broadened to include chemical and physical causes of disease or injury.

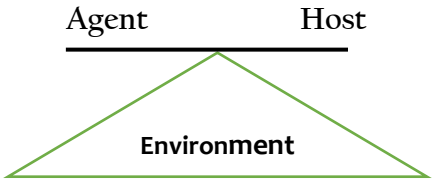
Host

- Refers to the human who can get the disease
- Is the living being that the bacteria, virus, protozoan, or other disease-causing microorganism normally resides in
- A variety of factors intrinsic to the host, sometimes called risk factors, can influence an individual's exposure, susceptibility, or response to a causative agent.
- Host factors - Are factors that influence individual's exposure, susceptibility, or response to a causative agent. Age, race, sex, socio-economic status and behaviors are some of the many host factors which affect a person's likelihood of exposure

Environment

- Refers to extrinsic factors that affect the agent and the opportunity for exposure
- Environmental factors include:
  - Physical factors such as geology and climate
  - Biologic factors such as insects that transmit the agent
  - Socioeconomic factors such as crowding, sanitation, and the availability of health services.





NOTE: Agents, host and environmental factors interrelate in a variety of complex ways to produce a disease in humans.

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2.2. Disease and risk factors

Diseases are categorized in to communicable and non-communicable:

- **Communicable diseases**
  - These are infectious diseases caused by microorganisms such as bacteria, viruses, parasites and fungi that can be spread, directly or indirectly, from one person to another.
  - A communicable disease is one that is spread from one person to another through a variety of ways that include:
    - Contact with blood and body fluids;
    - breathing inn an airborne virus; or
    - by being bitten by an insect
  - Common communicable diseases are:
    - **Influenza** - Influenza (also known as the flu) is a contagious respiratory illness caused by flu viruses. It can cause mild to severe illness, and at times can lead to death. The flu usually comes on suddenly.
    - **Measles** - Measles is a highly contagious virus that lives in the nose and throat mucus of an infected person. The symptoms of measles generally appear about seven to 14 days after a person is infected.
    - **Meningitis** - Meningitis is the inflammation of the protective membranes covering the brain and spinal cord. Meningitis can be caused by various pathogens with the majority being either a bacterial or viral infection of the fluid surrounding these areas.
    - **Ebola** - Ebola is a rare and deadly hemorrhagic fever caused by an infection from one of the Ebola virus strains. Ebola can cause disease in humans and non-human primates
    - **Mumps** - Mumps is a contagious disease caused by a virus. An outbreak is considered when three confirmed cases occur at any time of the year
    - **Tuberculosis** – This is an airborne disease caused by the bacteria, which are spread person-to-person through the air. This bacterium mainly affects the lungs, but may also affect other organs
    - **Covid-19** – this is a communicable disease caused by a new strain of coronavirus that causes illness. Originally identified in China in 2019 and became pandemic in 2020
  - Risk factors of communicable diseases are:
    - Crowded conditions
    - Lack of vaccination (poor immunity)
    - Poor nutrition and diet
    - Forced displacement
    - Poor quality shelter
    - Poor water, sanitation and hygiene conditions
    - Lack of health care facilities
    - Lack of adequate surveillance

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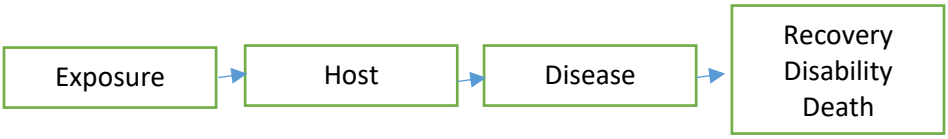
- **Non-communicable diseases**
  - Referred to as a "lifestyle" disease, because the majority of these diseases are preventable illnesses
  - These include diabetes, cancer, osteoporosis, stroke, and heart disease.
  - Almost four-fifths of all deaths from non-communicable diseases occur in developing countries
  - Risk factors of non-communicable are:
    - Lack of physical activities
    - Poor diet (high consumption of sugar, salt, saturated fats, and trans fatty acids)
    - Alcohol use
    - Metabolic/physiological changes (raised blood pressure, overweight/obesity, raised blood glucose and raised cholesterol)
    - Heart disease, stroke, cancer, diabetes and chronic lung disease

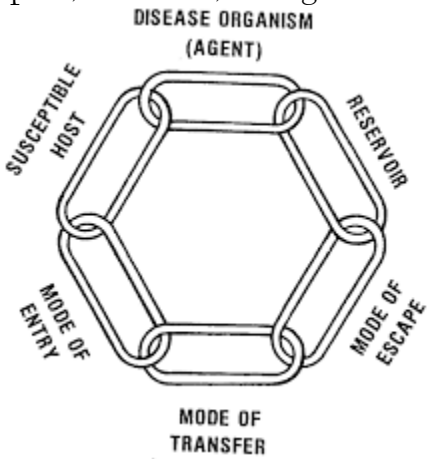
2.3. Natural history of disease

- This is the course a disease takes in individual people from its pathological onset until its eventual resolution through complete recovery or death
- Progress of a disease in an individual over time in the absence of intervention
- The process begins with exposure to factors capable of causing the disease
- Without medication intervention, the process ends with recovery, disability or death



2.4. Chain of infection

- Is also called a transmission cycle. It is made up of six different links: pathogen (infectious agent), reservoir, portal of exit, means of transmission, portal of entry, and the new host. Each link has a unique role in the chain, and each can be interrupted, or broken, through various means.



- Transmission cycle - occurs when the agent leaves its reservoir or host through a portal of exit, and is conveyed by some mode of transmission, and enters through an appropriate portal of entry to infect a susceptible host.
- **Reservoir** - is the habitat in which an infectious agent normally lives, grows and multiplies. This includes humans, animals and the environment.
- Two types of human reservoir are:
  - **Persons with symptomatic illness** - Symptomatic persons are usually less likely to transmit infection widely because their symptoms increase their likelihood of being diagnosed and treated.

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- **Carriers** - Asymptomatic carrier, a person or organism infected with an infectious disease agent, but displays no symptoms.

- **Environmental reservoir** - Plants, soil and water are also reservoir for some infectious disease. Many fungal agents such as those causing histoplasmosis live and multiply in the soil
- **Animal reservoir** - Infectious diseases that are transmittable under normal conditions from animal to humans are called zoonoses. Such disease includes brucellosis (cows and pigs), anthrax (sheep), plague (rodents), rabies (bats, dogs)
- **Portal of exit and entry** - Is the path by which an agent leaves the source host or enters. Tubercle bacilli and influenza viruses exit the respiratory tract; Schistosome through urine; cholera vibrio through feces etc.
- **Modes of transmission** - After an agent exits its natural reservoir, it may be transmitted to a susceptible host in numerous ways. These modes of transmission are classified as:
    - **Direct** (Direct contact and direct spread) - This is immediate transfer of the agent from a reservoir to a susceptible host by direct contact or droplet spread. It occurs through kissing, skin-to-skin contact. Also refer to contact with soil or vegetation harboring infectious organisms.
    - **Indirect** (Airborne, vehicle borne, Vector borne) – This can be mechanical or biologic. An agent is carried from a reservoir (a susceptible host) by suspended air particles or by animate (vector).
    - Most vectors are arthropods such as mosquitoes, fleas and ticks. This is in contrast of instances in which an agent undergoes part of its life cycle inside a vector before being transmitted to a new host.
    - When the agent undergoes changes within the vector, the vector is serving as both intermediate host and a mode of transmission. This type of direct transmission is called a biologic transmission
    - Vehicles that may directly transmit an agent include food, water, blood, handkerchiefs, bedding etc.

2.5. Determinants of disease

- Any of a group of variables or factors, that directly or indirectly influence the frequency or distribution of a disease
- These can be:
    - **Environmental factors** - the greatest burdens relate to unsafe drinking water, poor sanitation & hygiene, and pollution of the indoor and outdoor air
    - **Biological factors** – This includes agents, such as Bacteria, Fungi, Parasite and Virus
    - **Genetics factors** - inheritance plays a part in determining lifespan, healthiness and the likelihood of developing certain illnesses.
    - **Physical factors** - safe water and clean air, healthy workplaces, safe houses, communities and roads all contribute to good health. Employment and working conditions – people in employment are healthier
    - **Individual factors** - Diet, Physical activity, life style such taking alcohol, cigarette, and other drug use, and Hand washing

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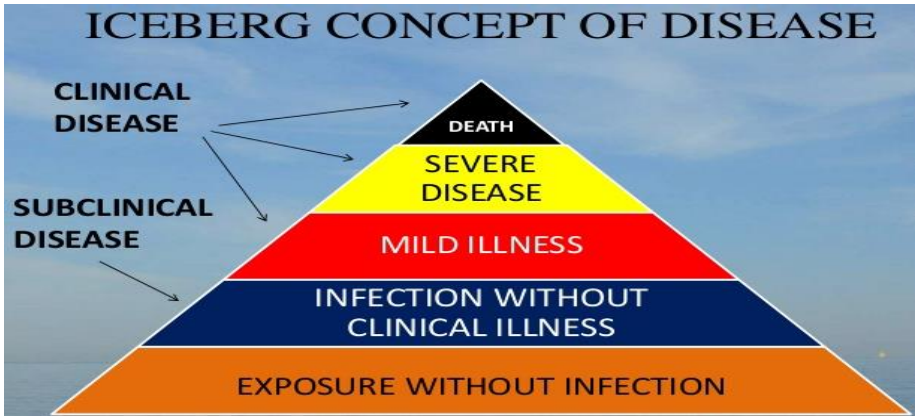
- **Social economic factors** - higher income and social status are linked to better health. The greater the gap between the richest and poorest people, the greater the differences in health.

2.6. Occurrences of diseases

- A disease and other health events do not occur randomly in a population, but are more likely to occur in some members of the population than others. This is because of risk factors that may not be distributed randomly in the population
- The amount of a particular disease that is usually present in a community is referred to as the baseline or endemic level of the disease. This level is not necessarily the desired level, but rather the observed level



- Some diseases are so rare in a given population that a single case warrants an epidemiologic investigation (e.g., rabies, plague, polio)
- Explanation of the Iceberg concept of diseases
  - From bottom – In a population, majority are exposed to diseases but don't get infected
  - Next up – the number that get infected with disease do not become sick
  - Next up – the number that become sick get healed without being noticed
  - Next up – the number of noticed sick people become severely sick and go to hospital
  - Next up – The number of severely sick people who goes to hospital eventually die

2.7. Methods of controlling communicable diseases

The following are methods of controlling communicable diseases:

- **Diagnosis and treatment** –This is identifying and treating cases as early as possible: Its purposes are:
  - Reducing the severity of the disease for the patient,
  - Avoiding progression to complications, disability and death
  - Reducing the risk of transmission to others.
- **Screening** - detection of an infection in an individual who does not show any signs or symptoms of the disease. It is carried out using specific tests called screening tests. It helps in:
  - Detecting an infection early
  - Organizing appropriate treatment so as to reduce complications
  - Preventing transmission to others

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- **Isolation** – separate patients from others to prevent transmission to healthy people. The isolation period lasts until the risk of transmission from the infected person has reduced or stopped.
- **Quarantine** – Restriction on movement of people who may have been exposed to a disease to prevent spread of disease or pest
- **Reporting** – Cases of communicable diseases should be reported to a nearby health center or health head office periodically, using the national surveillance guidelines
- **Terminating modes of transmission** - The measures that can be applied to interrupt transmission of infectious agents in water, food, other vehicles and by vectors i.e. sterilization, disinfection, boiling water etc.
- **Removing susceptible host** - help to protect the susceptible host either from becoming infected, or from developing the stage of infectious disease if they are exposed to the infectious agents i.e.
    - **Vaccination** - administration of vaccines to increase the resistance of the susceptible host against specific vaccine-preventable infections. For example, measles vaccination helps to protect the child from measles infection, and BCG vaccination gives some protection from tuberculosis
    - **Chemoprophylaxis** - drugs given to exposed and susceptible hosts to prevent them from developing an infection. For example, individuals from non-malarial areas who are going to a malaria endemic area can take a prophylactic drug to prevent them from developing the disease if they become infected with malaria parasites from a mosquito bite.
    - **Maintaining a healthy lifestyle** – Proper nutrition and exercise improves a person's health status, supports the effective functioning of their immune system, and increases resistance to infection
    - **Limiting exposure to reservoirs of infection** – Measures taken to decrease contact with reservoirs of infection include:
        - ✓ Condom use to prevent transmission of HIV and other STI
        - ✓ Use of insecticide treated nets (ITNs) over the bed at night, insect repellants and wearing protective clothing to prevent diseases transmitted by insect vectors.
        - ✓ Wearing surgical or very clean gloves and clean protective clothing while examining patients, particularly if they have wounds, or the examination involves the genital area.
        - ✓ Keeping personal hygiene i.e. hand washing

2.8. Level of disease prevention

- There are 3 levels of disease prevention: Primary, Secondary and Tertiary
- **Primary level** - It aims to prevent disease or injury before it ever occurs
    - This is done by preventing exposures to hazards that cause disease or injury, altering unhealthy or unsafe behaviors that can lead to disease or injury, and increasing resistance to disease or injury when exposure occur
    - Examples include:
        - legislation and enforcement to ban use of asbestos

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        - Education about healthy and safe habits (e.g. eating well, exercising regularly, not smoking)
        - Immunization against infectious diseases

- **Secondary level** - This level aims to reduce the impact of a disease or injury that has already occurred.
    - This is done by:
        - Detecting and treating disease or injury as soon as possible to halt or slow its progress
        - Encouraging personal strategies to prevent re-injury or recurrence
        - Implementing programs to return people to their original health and function
    - Examples include:
        - Regular exams and screening tests to detect disease in its earliest stages (e.g. mammograms to detect breast cancer)
        - Daily, low-dose aspirins and/or diet and exercise programs to prevent further heart attacks or strokes

- **Tertiary level** - This level aims to soften impact of an ongoing illness/injury that has lasting effects.
    - This is done by helping people manage long-term health problems and injuries (e.g. chronic diseases, permanent impairments) in order to improve as much as possible their ability to function, their quality of life and their life expectancy.

    - Examples include:
        - Cardiac or stroke rehabilitation programs, chronic disease management programs (e.g. for diabetes, arthritis, depression, etc.)
        - Support groups that allow members to share strategies for living well
        - Vocational rehabilitation programs to retrain workers for new jobs when they have recovered as much as possible.

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Unit 3. Measures of morbidity and mortality in epidemiology

3.1. Sources of epidemiological data

- Epidemiologists use primary and secondary data sources to calculate rates and conduct studies.
- Primary data:
  - This is the original data collected for a specific purpose by or for an investigator
  - For example, an epidemiologist may collect primary data by interviewing people who became ill after eating at a restaurant in order to identify which specific foods were consumed.
  - Collecting primary data is expensive and time-consuming, and it usually is undertaken only when secondary data is not available.
- Secondary data
  - This is data collected for another purpose by other individuals or organizations.
  - Examples of sources of secondary data that are commonly used in epidemiological studies include:
    - Birth and death certificates,
    - Population census records,
    - Patient medical records,
    - Disease registers,
    - Insurance claim forms and billing records,
    - Public health department case reports,
    - Surveys of individuals and households.

3.2. Morbidity measurements

- **Incidence**
  - Are counts of new cases
  - **Incidence risk** is a measure of disease occurrence over a defined period of time. It is a proportion, therefore takes values from 0 to 1 (0% to 100%).
  - **Incidence rate** takes into account the time an individual is at risk of disease. It is not a proportion since it defines the number of cases per animal or farm time at risk.

- **Prevalence**
  - Are counts of new and old cases of a disease that are present in a particular population at a given time
  - It is derived by comparing the number of people found to have the condition with the total number of people studied, and is usually expressed as a fraction (i.e. the number of cases per 10,000 or 100,000 people)
  - The following are types of prevalence:
    - **Point prevalence** is the proportion of a population that has the characteristic at a specific point in time.
    - **Period prevalence** is the proportion of a population that has the characteristic at any point during a given time period of interest. "Past 12 months" is a commonly used period.

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- **Lifetime prevalence** is the proportion of a population who, at some point in life has ever had the characteristic.

3.3. Mortality measurements

- **Crude death rate** is the number of deaths (from all causes) per 1000 estimated midyear population in a given place
- **Case fatality rate** – indicates killing power of a disease (e.g. number of death due to a specific disease divide by total number of same cases expressed in percentage)
- **Specific death rate** is number of death in a group during the year per 1000 midyear population. This can be Age Specific Death Rate or Disease Specific Death Rate
- **Survival rate** – proportion of survivors in a group studied and followed in a period of time. It shows the efficiency of treatment.
- **Infant mortality rate** – number of infants dying in a year per 1000 live births in a given geographical area
- **Child mortality rate** – number of deaths of children 1-4 years per 1000 population of the same age in a given year and in a given geographical area
- **Neonatal mortality rate** – number of infants dying within 4 weeks or up-to 28 days of life per 1000 live births in a year and in a geographical area
- **Early neonatal mortality rate** – number of infants dying within the first 7 days of life per 1000 live births in a year and in a geographical area
- **Perinatal mortality rate** – number of still births/late foetal death (after 28 weeks of gestation) plus deaths within 1st week of life in a year within a geographical area per 1000 births (live and still)
- **Maternal mortality rate** – number of death of women (while pregnant or within 42 days of termination of pregnancy) from any cause related to pregnancy or child bearing and child birth per 100,000 live birth in a given year
- **Still birth** – death of fetus after completing 28 weeks and till the time of birth

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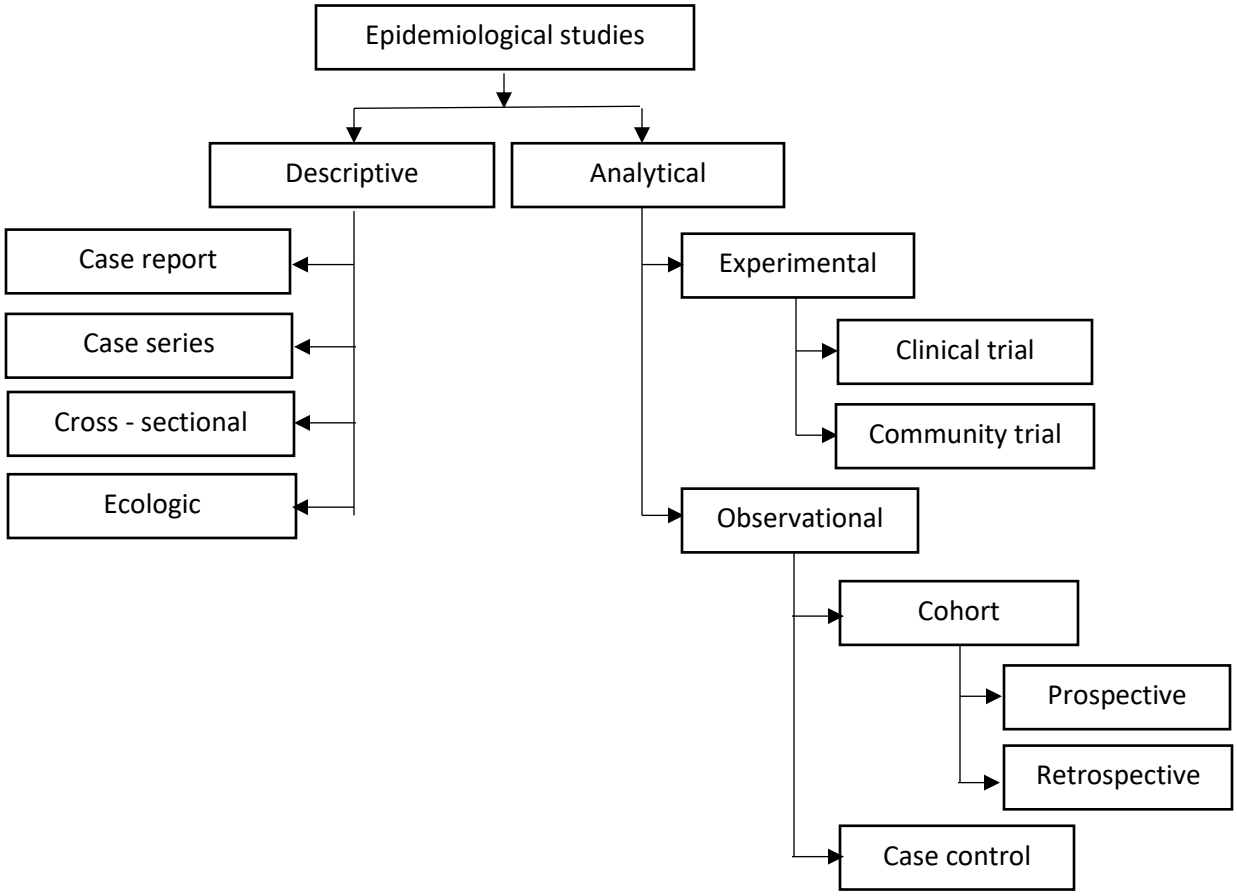
Epidemiology II (60 hours)

Unit 1. Epidemiological research designs

1.1. Summary diagram showing types of epidemiology



1.2. Descriptive

- Also called observational studies
- In this type of epidemiology, the disease distribution in a community is described in terms of Time, Place and Person
- Is used when little is known about the issue or disease and relies on pre-existing data
- Its purposes are:
    - Provides a systematic approach for dissecting a health problem into its component parts.
    - Ensures that you are fully versed in the basic dimensions of a health problem.
    - Identifies populations at increased risk for the health problem under investigation.
    - Provides timely information for decision-makers, the media, the public, and others about ongoing investigations.
    - Supports decisions for initiating or modifying control and prevention measures.
    - Measures the progress of control and prevention programs.
    - Enables generation of testable hypotheses regarding the etiology, exposure mode, control measure effectiveness, and other aspects of the health problem.
    - Helps validate the eventual incrimination of causes or risk factors.

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- Starts with no hypothesis regarding cause-effect relationship, and ends with development of possible hypothesis regarding cause and effect relationship but does not confirm or reject such hypothesis
- The procedure in descriptive epidemiology includes:
  - ✓ Define the population to be studied
  - ✓ Define the disease under study
  - ✓ Describe the disease by time, place, person
  - ✓ Develop measurements of the disease
  - ✓ Compare results with known indices
  - ✓ Formulate an etiological hypothesis

- The following are types of descriptive epidemiology
  - ✓ **Case report** – this is a detailed report of the symptoms, signs, diagnosis, treatment, and follow-up of an individual patient. They may contain a demographic profile of the patient, but usually describe an unusual or new occurrence.
  - ✓ **Case control** - is designed to help determine if an exposure is associated with an outcome (i.e., disease or condition of interest). A case-control study is always retrospective because it starts with an outcome then traces back to investigate exposures.
  - ✓ **Case series** – Patients with a particular condition or disease are sampled. It gathers experience of patients with similar disease
  - ✓ **Cross sectional** – It is also called prevalence study. Information is collected from a population in a point of time. For example, prevalence of respiratory conditions among furniture makers
  - ✓ **Ecological** – the incidence of a disease in one geographical area is compared to that of another area. For example, Cancer mortality in areas with hazardous waste site as compared to areas without waste sites

1.3. Analytical

- It relies on new data and is used when insight about issue or disease is available
- It starts by defining hypothesis regarding an exposure possibly causing an outcome, and ends with confirming or rejecting the hypothesis with which it started
- The following are 2 types of analytical epidemiology
  - ✓ **Experimental** - Deliberate intervention is made and the effect of such intervention is observed. There are three main types of experimental studies.
    - ***Randomized controlled trials (RCT)*** – uses patients as units of study, mostly for assessing a new therapy
    - ***Field trials*** – uses healthy individuals as units of study, mostly for assessing preventive agents like vaccines
    - ***Community trials*** – uses communities as units of study for assessing community based interventions like health status in villages with a training program versus in those without

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Unit 2. Measurements in epidemiological studies

2.1. Common measurements

- Data (information) are the building blocks of epidemiological research.
- Careful measurement, recording and handling of information are crucial to the research process.
- Measures of disease frequency including:
  - Prevalence
  - Incidence
  - Calculation of person-time at risk
  - Issues in defining the population at risk
  - The relationships between incidence and prevalence
  - Commonly used measures of disease frequency
- Measures of health outcome:
  - Morbidity
  - Mortality
  - Infectious disease incidence
  - Birth defects
  - Disability
  - Injuries
  - Vaccine efficacy
  - Utilization of hospital services
- Measures of effect

$$\textbf{Risk Ratio} = \frac{\text{Risk (cumulative incidence) in exposed group}}{\text{Risk (cumulative incidence) in unexposed group}}$$

$$\textbf{Rate Ratio} = \frac{\text{Incidence rate in exposed group}}{\text{Incidence rate in unexposed group}}$$

$$\textbf{Odds Ratio} = \frac{\text{Odds of disease in exposed group}}{\text{Odds of disease in unexposed group}}$$

2.2. Risk, Rates, ratio and proportions

Risk

- Risk is the proportion of individuals in a population (initially free of disease) who develop the disease within a specified time interval.
- Incidence risk is expressed as a percentage (or if small as per 1000 persons)

Rates

- Measure of the frequency with which an event occurs in a defined population over a specified period of time.
- Are useful for comparing disease frequency in different locations, at different times, or among different groups of persons with potentially different sized populations; that is, a rate is a measure of risk.

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- Describes how quickly disease occurs in a population, for example, 70 new cases of breast cancer per 1,000 women per year.

Ratio

- A comparison of any two values.
- It is calculated by dividing one interval- or ratio-scale variable by the other
- A ratio can describe the male-to-female ratio of participants in a study, or the ratio of controls to cases (e.g., two controls per case).
- As an analytic tool, ratios can be calculated for occurrence of illness, injury, or death between two groups

Proportion

- A comparison of a part to the whole where numerator is included in the denominator
- Describes what fraction of clinic patients tested positive for HIV, or what percentage of the population is younger than 25 years of age.
- A proportion may be expressed as a decimal, a fraction, or a percentage.
- Are used as descriptive measures. For example, one could calculate the proportion of persons enrolled in a study among all those eligible ("participation rate") OR the proportion of children in a village vaccinated against measles.
- Describe the amount of disease that can be attributed to a particular exposure. For example, 90% of the lung cancer cases that occur are attributable to cigarette smoking.

2.3. Reliability

- Reliability in statistics is the overall consistency of a measure. It is the degree to which the result of a measurement can be depended on to be accurate. OR the degree to which research method produces stable and consistent results
- A measure is said to have a high reliability if it produces similar results under consistent conditions OR a specific measure is considered to be reliable if its application on the same object in number of times produces the same results.
- The following are types of reliability:
    - **Test-retest** reliability - measures the consistency of results when you repeat the same test on the same sample at a different point in time. You can improve test reliability through:
        - When designing a questionnaire, formulate questions, statements and tasks in a way that won't be influenced by the mood or concentration of participants.
        - When planning your method of data collection, try to minimize the influence of external factors, and make sure all samples are tested under the same conditions.
        - Take in to account any change that is likely to occur during the assessment
    - **Interrater** reliability (also called inter-observer reliability) measures - the degree of agreement between different people observing or assessing the same thing. It is used when data is collected by researchers assigning ratings, scores or categories to one or more variables. It can be improved by:

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        - Clearly define your variables and the methods that will be used to measure them.
        - Develop detailed objective and criteria on how the variables will be rated, counted or categorized.
        - If multiple researchers are involved, ensure that they all have exactly the same information and training.

    - **Parallel forms** reliability measures - the correlation between two equivalent versions of a test. You use it when you have two different assessment tools or sets of questions designed to measure the same thing. It can be improved by:
        - Ensuring that all questions or test items are based on the same theory and formulated to measure the same thing.
    - **Internal consistency** measures - the correlation between multiple items in a test that are intended to measure the same construct.

2.4. Observer variation

- Failure by the observer in a study or test to measure accurately, resulting in error
- The following are subject variations
    - ***Inter-observer variation*** - is the amount of variation between the results obtained by two or more observers examining the same subject
    - ***Intra-observer variation*** - is the amount of variation one observer experiences when observing the same subject more than once

2.5. Subject variation

- This refers to the variation seen in a group of subjects which are all treated the same way.
- It is exhibited by a single person when given the same dose of a drug over repeated administrations.

2.6. Variability

- Variation means that when we measure something over-and-over again, we get a different result, and therefore, we cannot predict the outcome of any future observation.
- This is the extent to which data in a statistical distribution or data set differ from each other
- There are four measures of variability:
    - **Range**- the difference between the lowest and highest values.
    - **Interquartile range** - is a measure of variability, based on dividing a data set into quartiles. Quartiles divide a rank-ordered data set into four equal parts. The values that divide each part are called the first, second, and third quartiles; and they are denoted by Q1, Q2, and Q3, respectively. The interquartile range is equal to Q3 minus Q1.
    - **Variance** - measures how far a set of numbers are spread out from their average value
    - **Standard deviation** – Measures exact distances from the mean. It is calculated by getting the square root of the variance

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2.7. Test bias and confounding

- This is a situation in which the effect or association between an exposure and outcome is distorted by the presence of another variable.
    - **Positive** confounding (when the observed association is biased away from the null)
    - **Negative** confounding (when the observed association is biased toward the null)
- Test bias refers to the differential validity of test scores for groups (e.g., age, education, culture, race, sex).
- Bias is a systematic error in the measurement process that influences scores for identified groups

2.8. Error and systematic error

- A systematic error is where the same error is present in all readings
- It is predictable and generally constant or proportional to the true value

2.9. Repeatability

- This is a measure OR ability of the method to generate similar results for multiple observations of the same sample

2.10. Validity

- Validity is the extent to which an instrument, such as a survey or test, measures what it is intended to measure (also known as internal validity).
- This is important if the results of a study are to be meaningful and relevant to the wider population.
- There are four main types of validity:
    - **Construct validity** –
      The extent to which the instrument specifically measures what it is intended to measure, and avoids measuring other things. For example, a measure of intelligence should only assess factors relevant to intelligence and not, for instance, whether someone is a hard worker. **Does the test measure the concept that it's intended to measure?**
    - **Content validity** –
      Describes whether an instrument is systematically and comprehensively representative of the characteristics it is measuring. For example, a questionnaire aiming to score anxiety should include questions aimed at a broad range of features of anxiety. **Is the test fully representative of what it aims to measure?**
    - **Face validity** –
      The degree to which a test is subjectively thought to measure what it intends to measure. In other words, does it "look like" it will measure what it should do. The subjective opinion for face validity can come from experts, from those administering the instrument, or from those using the instrument. **Does the content of the test appear to be suitable to its aims?**
    - **Criterion validity** –
      Involves comparing the instrument in question with another criterion which is taken to be representative of the measure. This can take the form of:
        - **Concurrent** validity (where the instrument results are correlated with those of an established gold standard)

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- **Predictive** validity (where the instrument results are correlated with future outcomes, whether they be measured by the same instrument or a different one). **Do the results correspond to a different test of the same thing?**

2.11. Sensitivity and specificity

- **Sensitivity test** - is the ability of a test to correctly identify those with the disease (true positive rate)
- **Specificity test** - is the ability of the test to correctly identify those without the disease (true negative rate).

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- Calculations
  - Point prevalence = Number of cases in a defined population at one point in time ÷ Number of persons in a defined population at the same point in time
  - Incidence risk = Number of new cases of diseases in a specified period of time ÷ Number of disease-free persons at the beginning of that time period
  - Incidence rate = Number of new cases of disease in a given time period ÷ Total person-time at risk during the study period

$$\textbf{Age/sex specific mortality rate} = \frac{\text{number of deaths in year in specific age/sex group}}{\text{mid-year population of age or sex group}} \times 1000$$

$$\textbf{Birth rate} = \frac{\text{number of births in a year}}{\text{mid-year population}} \times 1000$$

$$\textbf{Fertility rate} = \frac{\text{number of live births in year}}{\text{mid-year population of women aged 15-44}} \times 1000$$

$$\textbf{Infant mortality rate} = \frac{\text{number of deaths in year <1 year of age}}{\text{number of live births in year}} \times 1000$$

$$\textbf{Perinatal mortality rate} = \frac{\text{number of stillbirths + deaths <7 days in a year}}{\text{total number of births (live + still) in a year}} \times 1000$$

$$\textbf{Neonatal mortality rate} = \frac{\text{number of deaths in year < 28 days of age}}{\text{number of live births in a year}}$$

$$\textbf{Case fatality rate (\%)} = \frac{\text{number of deaths in year (from specific disease}}{\text{number of cases of that disease in a year}} \times 100$$

$$\textbf{Proportional mortality rate due to TB} = \frac{\text{total number of deaths due to TB}}{\text{total number of deaths due to all causes}} \times 100$$

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Unit 3. Disease surveillance

3.1. Definition of terms

- **Disease** - A disease is an abnormal condition that negatively affects the function of part or organism, and that is not due to any external injury
- **Outbreak** - a sudden increase in occurrences of a disease in a particular time and place
- **Investigation** - study of facts that are then used to inform decision
- **Outbreak investigation** – identification of source of disease in order to establish control and to institute measures for preventing future episodes of disease
- **Surveillance** - is the ongoing systematic collection, analysis and interpretation of data for planning, implementation, and evaluation
- **Disease surveillance** - information-based activity involving the collection, analysis and interpretation of data to evaluate the effectiveness of control and preventative health measures

3.2. Outbreak (epidemic) detection

- Detecting an outbreak is the first step in investigating an outbreak
- Outbreaks are detected by:
  - Using surveillance methods
  - Formal reports of illnesses
  - Informal reports of illnesses
- If the index exceeds a critical value, then an epidemic will begin
- The index case is the patient in an outbreak who is first noticed, and who makes people aware that an outbreak might be emerging
- Epidemic is detected if the sensitivity, specificity and positive predictive values show indications of an epidemic

3.3. Types of epidemics

- **Point source**
  - An epidemic in which all cases are infected at the same time, usually from a single source or exposure.
  - A common example will be food from a single meal at an event.
  - In this type of outbreak, the number of cases rises rapidly to a peak and decline gradually.
- **Continuous source**
  - An epidemic in which the causal agent is continuously infecting people who come into contact with it, over an extended period of time.
- **Person-to-Person**
  - Is also called Propagated
  - The causal agent is spread through person to person.
  - For this reason, the epidemic may last longer and is more complicated to contain

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3.4. Investigation of epidemics/outbreaks

- Objectives of investigating epidemics
  - Define magnitude of the epidemic (When, Whom, Where)
  - Determine factors responsible for the epidemic (Why)
  - Identify course, source of infection and modes of transmission (How)
  - Implement control and preventive measures at the beginning of epidemic

- Ten steps of epidemic investigation
  1. **Prepare for field work** – by addressing scientific and investigative issues (review literatures about the issue), and management and operational issues (team formation and planning for logistics)
  2. **Establish the existence of an outbreak** – by checking reports if there are more cases than expected
  3. **Verify the diagnosis** - by reviewing the clinical findings and laboratory results
  4. **Construct a working case definition** - by designing standard set of criteria for deciding whether an individual should be classified as having the health condition of interest
  5. **Find cases systematically and record information** – by conducting passive surveillance and active surveillance to collect Identifying information, Demographic information, Clinical information, Risk factor information, Reporter information
  6. **Perform descriptive epidemiology** – by characterizing the occurrence by time, place, and person
  7. **Develop hypotheses** - by addressing the source of the agent, the mode (and vehicle or vector) of transmission, and the exposures that caused the disease. The hypotheses should be testable, since evaluating hypotheses is the next step in the investigation
  8. **Evaluate hypotheses epidemiologically** – by 1) comparing the hypotheses with the established facts or by using analytic epidemiology to quantify relationships and assess the role of chance; 2) using a combination of environmental evidence, laboratory science, and epidemiology
  9. **Implement control and prevention measures** – by 2) blocking the mode of transmission and 2) interrupting direct transmission through isolation, counselling and treatment
  10. **Initiate or maintain surveillance** – Monitor effectiveness of the intervention through routine collection, analysis and interpretation of data
  11. **Communicate findings** – by giving oral briefing for local authorities and written reports.

3.5. Control of epidemics

- In order to control epidemics, we must have information on:
  - Causative organism
  - Dynamics of disease transmission
  - Modes of transmission
  - Root of transmission

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- Apply the 3 dynamics of disease prevention
    - Removal of source of infection
    - Prevent transmission
        - Isolation of infected cases
        - Hand washing and personal protective measures
        - Proper disposal of waste
        - Screening of suspected cases
        - Health education
        - Quarantine of migrated cases
        - Increase resistance through immunization
    - Vector control measure
        - Prevent breeding of vectors
        - Destruction of adult vectors through insecticide and pesticide
        - Personal protection from bites
        - Improve environmental sanitation

3.6. Purposes of surveillance

- Provide information for action
- Monitor disease trend
- Detecting outbreaks
- Determine program and resource needs
- Stimulate research
- Trigger disease control activities
- Reduce infection rate
- Establish epidemic baseline
- Evaluate disease prevention control measures
- Defend claims for malpractice
- Assess public health status
- Estimating the magnitude of a health problem

3.7. Types of surveillance

- **Passive**
    - Routine reporting system done at hospitals to gather disease data from all potential reporting health care workers.
    - Health authorities do not stimulate reporting by reminding health care workers to report disease nor providing feedback to individual health workers
- **Active**
    - Provides stimulus to health care workers in the form of individual feedback or other incentives.
    - Frequency of reporting by individual health workers is monitored
    - Health workers who consistently fail to report or complete the forms incorrectly are provided specific feedback to improve their performance
- **Sentinel surveillance**
    - Instead of attempting to gather surveillance data from all health care workers, a sentinel surveillance system selects, either randomly or intentionally, a small group of health workers from whom to gather data.

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- Sentinel surveillance also requires more time and resources, but can often produce more detailed data on cases of illness because the health care workers have agreed to participate and may receive incentives.

3.8. Components of surveillance

- Data collection
- Data analysis
- Dissemination
- Response

3.9. Sources of health surveillance data

- Health facilities registers
- Patient treatment notes
- Summary reports
- Notifications
- Activity reports
- Review of surveillance data
- Vital statistics and verbal autopsy – data on events in life e.g. birth, death, marriage
- Demographic surveillance sites
- Disease registries
- Health surveys
- Sentinel systems
- Administrative data
- Census

3.10. Strategies of surveillance

- Carry out surveillance that identifies and drives the promotion of health and well-being
- Strengthen surveillance capacity and transform data into quality information and knowledge
- Creating surveillance system that fits into existing integrated disease surveillance
- Continuous improvement in surveillance methods through learning from local, national and international experience
- Automation of routine collection, analysis and reporting of surveillance information
- Building surveillance systems that are adaptable and respond to changing needs in a timely manner
- Ensuring mechanisms are in place to be aware of new and re-emerging infectious diseases and possible environmental health hazards

3.11. Steps in surveillance

- Selection of topic and measure
- Identification of data sources and creation of indicators
- Data collection
- Data integration
- Data analysis
- Interpretation of results
- Development of surveillance report
- Dissemination of report

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3.12. Attributes of a good surveillance system

- **Simplicity** of the system structure and operation
- **Flexibility** to look at questions posed by research and accommodate changes or reporting definitions
- **Data quality** for complete and valid data
- **Acceptability** to generate willingness to participate in the system
- **Sensitivity** for capturing all monitoring trends
- **Predictive** value positive of capturing true cases
- **Representativeness** of the information for individuals in the study
- **Timeliness** of process from collecting information to decision making
- **Stability** to reliably operate and provide information when called upon
- Collecting **specific** information that is required to achieve objectives for disease control

3.13. Disease threshold for epidemics

- A threshold is a marker that alerts public health officials to take action.
- It uses past data to decide if an event is abnormal in order to identify possible outbreaks using surveillance data.
- Although there are internationally accepted signals that constitute an alert for some conditions, there are no globally fixed criteria for all diseases to guide outbreak detection.
- Diseases under elimination, e.g. a single cases of poliomyelitis, Guinea worm or ebola constitutes an alert.
- Characteristics and settings of a population, such as overcrowded camp settings of displaced populations at risk following man-made or natural disasters. In such settings a single case of meningitis constitutes an alert.
- For high frequency, less severe infectious diseases, several cases are normally expected in endemic settings and a single case is not abnormal. However, we would want to know and investigate further when a disease occurs at levels exceeding expectation (or "baseline") by a significant amount.
- A threshold helps us decide if what we are seeing is abnormal and needs to be investigated further. Typhoid, for example, is a high-frequency disease in developing countries. Every reported case does not necessarily require a public health response.
- Only when cases are clustering within a defined area or there is a substantial increase in the occurrence of cases is follow-up required.

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3.14. Disease surveillance in Kenya

- The IDSR is defined as a strategy for multi-disease surveillance of selected priority diseases or conditions
- It was established in Kenya in 2002 with the following purposes:
  - To ensure the availability of guidelines for IDSR
  - To provide necessary technical support for strengthening national communicable disease surveillance systems
  - Link the regional IDSR and the national IDSR
  - Establishes links with national Centres of Excellence (CoE), regional and global networks for disease surveillance
  - Envisages all surveillance activities in a country as a common public service.
  - Links the community, health facility, county and national levels in disease control and prevention
- The IDSR priority diseases are:
  - Cholera
  - Dysentery
  - Plague
  - Yellow fever
  - Typhoid fever
  - Meningococcal Meningitis
  - Measles
- Disease earmarked for eradication and elimination are:
  - Leprosy
  - Poliomyelitis
  - Neonatal tetanus
  - Dracunculiosis
- Diseases of public health importance are:
  - Malaria
  - New AIDS cases
  - Tuberculosis
  - Childhood pneumonia
  - Childhood diarrhea
  - STI
- Challenges faced by IDSR in Kenya:
  - Linkage to action is weak at lower levels due to lack of capacity for data analysis
  - Weak reporting & communication system from health facilities to county/central surveillance unit
  - Weak laboratory capacity & network
  - Lack of quarantine & adequate isolation facilities
  - Insufficient early warning systems for epidemic preparedness & response
  - Little or Non-involvement of the communities in surveillance
  - Limited financial resources to strengthen surveillance & to support epidemic response

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Unit 4. Management of epidemiological data

4.1. Epidemiological data

- Epidemiology is the study of the distribution and determinants of health related states or events within a specified population
- Its purpose being to inform decisions about the control of health problems, it relies much on primary data
- Primary data is the original data collected for a specific purpose by or for an investigator. For example, an epidemiologist may collect primary data by interviewing people who became ill after eating at a restaurant in order to identify which specific foods were consumed.

4.2. Overview of survey

- A Survey is defined as a research method used for collecting data from a pre-defined group of respondents to gain information and insights on various topics of interest.
- Surveys can be carried out in many ways depending on the methodology chosen and the objectives to be achieved.
- The Survey method is the technique of gathering data by asking questions to people who are thought to have desired information.
- A formal list of questionnaire is prepared.
- Surveys can be divided into two broad categories:
    - Questionnaires - are usually paper-and-pencil instruments that the respondent completes.
    - Interviews - are completed by the interviewer based on the respondent says
- There are three main types of longitudinal survey:
    - **Trend surveys** - In a trend study we compare sample surveys describing the same population at two or more points in time
    - **Panel surveys** - is a type of longitudinal survey. It can be a qualitative or quantitative survey used to measure people's behavior.
    - **Cohort surveys** - is a longitudinal study that samples a cohort, performing a cross-section at intervals through time. It is a type of panel study where the individuals in the panel share a common characteristic

4.3. Major steps of survey

1. Determine Decisions Regarding Control Measure Implementation
2. Define The Investigation's Objectives And Determine Data Needed
3. Develop a Study Protocol
    - A field investigation protocol does not have to be long, but it must include the following:
    - Investigation objectives.
    - Study design (e.g., cohort study, case–control study).
    - Study population, case definition, sample size, and selection.
    - Data collection procedures, variables to be collected, procedures to safeguard participants.

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- Data security, privacy, confidentiality, information technology controls.
- Analysis plan.
- Logistics, including budget, personnel, and timeline.
- Legal considerations, including statutes, rules, and regulations.

4. Identify Possible Data Sources
    - Mortality Statistics
    - Notifiable Diseases Reporting
    - Laboratory Data
    - Ongoing Population Surveys
    - Environmental Exposure Data
5. Determine Data Collection Method
6. Develop the Questionnaire or Survey Instrument
    - **Close-ended questions** - These questions ask participants to choose from predetermined response categories.
    - **Open-ended questions** - These questions enable participants to answer in their own words and can provide rich information about new topics
    - Precoded, open-ended questions. These questions can be used on interviewer-administered surveys. They enable participants to answer unprompted, but the interviewer selects from precoded response categories.
7. Calculate the Sample Size and Select the Sample
8. Collect the Data
    - Having a written data collection section as part of the overall study protocol is essential.
    - This protocol of data collection can include the following:
        - Introductory letter to participants.
        - Introductory script for interviewers.
        - Instructions for recruiting and enrolling participants in the survey
        - Instructions on conducting the interviews i.e. , importance of reading the questions verbatim, term definitions, the pace of the interview, answers to frequently asked questions, and ways to handle urgent situations.
        - Instructions related to protection of participants (e.g., maintaining confidentiality, data security).
9. Data analysis
    - For proper data analysis:
        - Monitor data quality at all stages by following indicators
        - Follow proper data entry protocol
        - Compile and clean data from different sources
        - Conduct proper analysis of weekly aggregate data and disease outbreak data, including
        - Tabulation and indicator calculation
        - Highlight and summarize key points like aberrations from the normal pattern, start and end date of possible outbreaks, clustering, and other trends and patterns in the data.
    - Analyze the distribution of a disease in terms of person, time and place

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